

Continuous code of inspection	Continuous code of inspection allows performing static code analysis and highlights bugs, vulnerabilities, and other code issues related to standards. Quality profiles and quality gate can be integrated with a continuous integration server. In short, static code analysis (SCA) helps analyse the code without running it. The benefits of implementing continuous code inspection practice in the organisation are: • SCA tools are open source, affordable, or available with SDKs • Automated code analysis • Repeatable • Detailed analysis of the AS-IS scenario of code • Customisable • No human intervention except initial configuration for code analysis • Huge rule set and efficient scanning of a large code-base The tools related to continuous code of inspection are: • SonarQube • Android Lint • SwiftLint • Checkstyle																		
Continuous integration	Continuous integration is the DevOps practice to integrate new feature implementation or a bug fix in the existing distributed version control systems such as Git. This triggers code quality checks, unit test execution, and build. Continuous integration is easy to implement; hence for DevOps starters, it works as a solid base on which the rest of the DevOps practices can stand. The benefits of implementing continuous integration practices in the organisation are: • Reduce repetitive manual processes • Improve quality • Increase confidence • Faster detection of failures • Less effort in the resolution of errors in the early stages of software development • Focus on new feature implementations and bug fixes • Collective responsibility, accountability, and ownership • Reduce risks The tools related to continuous integration are: • Microsoft Azure DevOps • Jenkins • Atlassian Bamboo • TeamCity																		
Continuous delivery and deployment	The objective of continuous delivery and deployment is to deploy an application in the Dev, test, UAT, and production environments in an automated manner. This helps in incremental releases after short spans of development (or sprints in agile terms). <table><tr><td></td><td>Continuous delivery</td><td>Continuous deployment</td></tr><tr><td>Definition</td><td>Continuous delivery is about the confidence and the capabilities that every change is ready to be deployed to production immediately if required</td><td>Continuous deployment is about the confidence and the capabilities that every change has been deployed in the production environment</td></tr><tr><td>Environment</td><td>Production-like environments such as Dev, test, SIT, UAT, and so on.</td><td>Production</td></tr><tr><td>Feasible</td><td>Yes</td><td>Several organisations and projects prefer not to deploy all changes directly to production environments</td></tr><tr><td>Approvals</td><td>Manual/stakeholder approval is needed</td><td>Manual/stakeholder approval is not needed</td></tr><tr><td>Relation</td><td>Continuous delivery doesn't imply continuous deployment</td><td>Continuous deployment implies continuous delivery</td></tr></table>		Continuous delivery	Continuous deployment	Definition	Continuous delivery is about the confidence and the capabilities that every change is ready to be deployed to production immediately if required	Continuous deployment is about the confidence and the capabilities that every change has been deployed in the production environment	Environment	Production-like environments such as Dev, test, SIT, UAT, and so on.	Production	Feasible	Yes	Several organisations and projects prefer not to deploy all changes directly to production environments	Approvals	Manual/stakeholder approval is needed	Manual/stakeholder approval is not needed	Relation	Continuous delivery doesn't imply continuous deployment	Continuous deployment implies continuous delivery
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